Degree Heterogeneity, Eigenvector Localization, and Regime Interactions in Ordinary and Normalized Spectral Graph Clustering

Lakshya
Department of Computer Science and Engineering, Indian Institute of Technology Delhi
Repository: github.com/lakshya07iitd-ai/spectral-graph-clustering

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Abstract

Spectral graph clustering partitions networks using low-frequency eigenvectors of graph Laplacian operators, yet practitioners face a fundamental choice between the ordinary combinatorial Laplacian $L = D - A$ and the symmetric normalized Laplacian $L_{\text{sym}} = D^{-1/2}LD^{-1/2}$. While asymptotic theory demonstrates that L_{sym} converges to a well-behaved compact integral operator under broad degree distributions whereas L possesses a continuous multiplication spectrum spanning $[d_{\min}, d_{\max}]$, empirical benchmarks on finite graphs often confound node-level degree dispersion with community-size skew and shifted inter-community edge mixing. This paper presents a controlled experimental investigation of ordinary and normalized spectral clustering implemented from scratch in Python and NumPy. First, we show mathematically that naive degree-corrected stochastic block models (DC-SBMs) reduce expected within-community edge density as degree variance increases due to self-loop exclusion and hub-pair probability clipping. We resolve this confound by introducing a Separation-Calibrated DC-SBM that enforces exact within-community propensity normalization and bisection-calibrated edge expectations ($\mathbb{E}[m_{\text{in}}] = 87.75$, $\mathbb{E}[m_{\text{out}}] = 24.00$ on $N = 30, K = 3$ graphs). Second, across 90 matched trials at baseline separation ($p_{\text{in}} = 0.65, p_{\text{out}} = 0.08$), paired Adjusted Rand Index (ARI) gain $\Delta\text{ARI} = \text{ARI}_{\text{norm}} - \text{ARI}_{\text{ord}}$ rises monotonically from $+0.0220$ ($p = 0.1718$, non-significant) under balanced degrees to $+0.0694$ ($p = 0.0307$) under moderate degree imbalance and $+0.0927$ ($p = 0.0089$) under high degree imbalance. Third, a two-way regime investigation across 450 graph instances reveals a strong interaction between degree heterogeneity (γ) and community mixing (p_{out}): when communities are strongly separated ($p_{\text{out}} = 0.02$), both operators achieve ARI $= 1.0000$, whereas under elevated mixing ($p_{\text{out}} = 0.22$) and high degree heterogeneity ($\gamma = 0.85$), ΔARI reaches $+0.5509$ (95% CI $[+0.462, +0.640]$, Wilcoxon $p = 1.73 \times 10^{-6}$). Mechanistic spectral diagnostics demonstrate that ordinary clustering failure is driven by peripheral eigenvector localization whenever the k-th Laplacian eigenvalue approaches the minimum vertex degree ($\lambda_3(L)/d_{\min} \to 1$), inflating the Inverse Participation Ratio (IPR(L) correlates with ARI_{ord} at $r = -0.7494$). Finally, we identify the sparse hub-periphery regime ($p_{\text{in}} = 0.32, \bar{d}_{\min} = 0.17$) where standard L_{sym} itself degrades to ARI $= 0.2134$ due to disconnected and pendant vertices, and verify that degree-regularized spectral clustering (L_τ) restores accuracy to ARI $= 0.7357$.

Keywords: Spectral Graph Clustering, Graph Laplacian, Normalized Cut, Degree-Corrected Stochastic Block Model, Eigenvector Localization, Eigengap, Degree Regularization.

1 Introduction

Community detection in undirected networks is a foundational problem in machine learning, network science, and combinatorial optimization [2, 7]. Given an undirected graph $G = (V, E)$ with non-negative adjacency matrix $A \in \mathbb{R}_{\geq 0}^{n \times n}$, spectral clustering embeds vertices into $\mathbb{R}^k$ using the bottom k eigenvectors of a graph Laplacian matrix and partitions the resulting embedding via k-means clustering [4,8]. Because spectral relaxations convert NP-hard discrete cut objectives into tractable symmetric eigenvalue problems [1, 7], they provide both mathematical transparency and strong empirical performance.

Despite identical computational complexity, two distinct Laplacian operators dominate the literature: the ordinary (unnormalized) Laplacian $L = D - A$, which relaxes RatioCut, and the symmetric normalized Laplacian $L_{\text{sym}} = D^{-1/2}LD^{-1/2}$, which relaxes Normalized Cut (Ncut) [7, 8]. On nearly regular graphs where all vertices share similar degrees, $D \approx \bar{d}I$ and the eigenvectors of L and L_{sym} span nearly identical subspaces. Real-world networks, however, exhibit pronounced degree heterogeneity, combining high-degree hub vertices with low-degree peripheral vertices inside the same community [3].

2 Research Motivation

While theoretical analyses establish that unnormalized spectral clustering can fail asymptotically when the minimum degree $d_{\min}$ overlaps the nontrivial community eigenvalues of L [9], finite-sample empirical studies frequently suffer from three methodological weaknesses:

1. **Confounded Graph Generation:** When synthetic graphs are generated via naive Degree-Corrected Stochastic Block Models (DC-SBMs) [3], increasing vertex propensity variance simultaneously alters within-community edge counts and community volume balance, confounding degree heterogeneity with community separation.
2. **Confounded Algorithmic Choices:** Standard implementations of normalized spectral clustering apply Ng–Jordan–Weiss (NJW) spherical row normalization [4], leaving unclear whether empirical gains stem from the normalized operator L_{sym} or from post-eigenvector row normalization.
3. **Single-Regime Overgeneralization:** Reporting a single benchmark setting obscures the boundary conditions where L performs comparably to L_{sym} as well as the sparse regime where L_{sym} itself breaks down [5].

This study addresses all three issues through a rigorously controlled empirical investigation built atop a verified from-scratch spectral graph library.

3 Research Questions and Hypotheses

Our primary research question asks: *How does within-community degree heterogeneity affect the relative community recovery accuracy of ordinary (L) and symmetric normalized (L_{sym}) spectral clustering when community separation and expected edge densities are held strictly constant, and how does this relationship vary across network mixing and sparsity regimes?*

We formulate and test three falsifiable hypotheses:

- **Hypothesis H1 (Monotonic Divergence at Fixed Separation):** At a fixed intermediate community separation where both methods succeed under homogeneous degrees, increasing within-community degree dispersion (measured by the degree coefficient of variation CV_d) degrades ordinary spectral clustering while leaving normalized spectral clustering stable, yielding a positive and increasing paired accuracy gap $\Delta\text{ARI} = \text{ARI}_{\text{norm}} - \text{ARI}_{\text{ord}}$.
- **Hypothesis H2 (Regime Interaction and Non-Uniformity):** The advantage of L_{sym} over L is not uniform across graph structures; it vanishes when inter-community mixing is sufficiently low ($\lambda_k(L) \ll d_{\min}$) and amplifies super-additively as inter-community mixing pushes $\lambda_k(L)$ toward $d_{\min}$.
- **Hypothesis H3 (Peripheral Localization vs. Sparse Breakdown):** Ordinary clustering failures under degree heterogeneity are driven by eigenvector localization onto low-degree peripheral vertices rather than hub dominance; conversely, in sparse hub-periphery graphs where $d_{\min} \to 0$, standard L_{sym} loses its advantage unless stabilized by degree regularization (L_τ).

4 Relevant Literature Review

Spectral graph partitioning originated with Fiedler's analysis of the second smallest eigenvalue $\lambda_2(L)$ (algebraic connectivity) and its eigenvector for two-way graph bisection [1]. Shi and Malik [7] demonstrated that minimizing raw cut or Ratio-Cut often isolates single low-degree vertices and introduced the Normalized Cut criterion, showing that its continuous relaxation is solved by the generalized eigenvalue problem $Lx = \lambda Dx$, equivalent to the standard eigenvalue problem for $L_{\text{sym}} = D^{-1/2}LD^{-1/2}$. Ng, Jordan, and Weiss [4] extended symmetric normalized spectral clustering to k communities by projecting the bottom k eigenvectors of L_{sym} onto the unit sphere via row normalization prior to k-means clustering.

From a statistical perspective, von Luxburg, Belkin, and Bousquet [9] proved a fundamental dichotomy: as sample size $n \to \infty$, $L_{\text{sym}} = I - D^{-1/2}AD^{-1/2}$ is a compact perturbation of the identity whose isolated bottom eigenvalues converge consistently, whereas $\frac{1}{n}L$ converges to a multiplication operator whose continuous essential spectrum spans the range of the degree function. Consequently, whenever a nontrivial cluster eigenvalue lies inside $[d_{\min}, d_{\max}]$, the corresponding sample eigenvector localizes around a single low-degree vertex [8,9].

To model real-world networks with broad degree distributions, Karrer and Newman [3] proposed the Degree-Corrected Stochastic Block Model (DC-SBM). Rohe, Chatterjee, and Yu [6] proved high-dimensional consistency of L_{sym} under dense SBMs using Davis–Kahan subspace perturbation bounds [10]. Subsequently, Chaudhuri et al. and Qin and Rohe [5] showed that under sparse DC-SBMs with very low-degree vertices, L_{sym} suffers from high variance on pendant and isolated vertices, motivating the regularized Laplacian $L_\tau = I - D_\tau^{-1/2}AD_\tau^{-1/2}$ with $D_\tau = D + \tau I$.

Distinction of Our Contribution: The theoretical properties of L, L_{sym}, and L_τ are established in the literature [3,5,9]. Our contribution is a rigorous finite-sample experimental methodology and mechanistic verification: we derive and correct the finite-graph self-loop and clipping confounds in DC-SBM benchmarking, decouple operator normalization from spherical row normalization via explicit ablations, and quantify how the spectral threshold ratio $\lambda_3(L)/d_{\min}$ and Inverse Participation Ratio (IPR) govern finite-graph failure modes.

5 Mathematical Background

Let $G = (V,E)$ be an undirected simple graph on $n = |V|$ vertices with symmetric adjacency matrix $A \in \{0,1\}^{n\times n}$ ($A_{ii} = 0$). The vertex degrees are $d_i = \sum_{j=1}^n A_{ij}$, forming $D = \text{diag}(d_1, \ldots, d_n)$. The ordinary graph Laplacian is:

$$L = D - A. \tag{1}$$

For any vector $x \in \mathbb{R}^n$, L satisfies the edge-based quadratic form identity:

$$x^T Lx = \frac{1}{2}\sum_{i=1}^n \sum_{j=1}^n A_{ij}(x_i - x_j)^2 \geq 0. \tag{2}$$

Thus L is symmetric positive semi-definite with eigenvalues $0 = \lambda_1(L) \leq \lambda_2(L) \leq \cdots \leq \lambda_n(L)$, and the multiplicity of the zero eigenvalue equals the number of connected components of G [8].

Defining the safe inverse square-root degree matrix $D^{-1/2} = \text{diag}(d_1^{-1/2}, \ldots, d_n^{-1/2})$ with $d_i^{-1/2} := 0$ whenever $d_i = 0$, the symmetric normalized Laplacian is:

$$L_{\text{sym}} = D^{-1/2}LD^{-1/2} = I_{\text{pos}} - D^{-1/2}AD^{-1/2}, \tag{3}$$

with eigenvalues $0 = \lambda_1(L_{\text{sym}}) \leq \lambda_2(L_{\text{sym}}) \leq \cdots \leq \lambda_n(L_{\text{sym}}) \leq 2$. Its quadratic form normalizes each vertex value by $\sqrt{d_i}$:

$$x^T L_{\text{sym}} x = \frac{1}{2}\sum_{i=1}^n \sum_{j=1}^n A_{ij}\left(\frac{x_i}{\sqrt{d_i}} - \frac{x_j}{\sqrt{d_j}}\right)^2. \tag{4}$$

For a two-way partition $(S, \bar{S})$, with $\text{cut}(S,\bar{S}) = \sum_{i\in S, j\in\bar{S}} A_{ij}$ and volume $\text{vol}(S) = \sum_{i\in S} d_i$, the Normalized Cut is:

$$\text{Ncut}(S,\bar{S}) = \frac{\text{cut}(S,\bar{S})}{\text{vol}(S)} + \frac{\text{cut}(S,\bar{S})}{\text{vol}(\bar{S})}. \tag{5}$$

While minimizing $\text{RatioCut}(S,\bar{S}) = \text{cut}(S,\bar{S})(1/|S| + 1/|\bar{S}|)$ relaxes to the Fiedler vector of L, minimizing $\text{Ncut}(S,\bar{S})$ relaxes to the second eigenvector of L_{sym} [7]. By the Davis–Kahan $\sin\Theta$ theorem [10], the stability of the k-dimensional

community subspace under edge perturbations scales inversely with the k-th eigengap:

$$\Delta\lambda_{k+1,k} = \lambda_{k+1} - \lambda_k, \tag{6}$$

which for $k = 3$ communities corresponds to $\lambda_4 - \lambda_3$ (`values[3] - values[2]` in zero-based array indexing).

6 Graph-Generation Methodology

Consider $N = 30$ vertices partitioned into $K = 3$ planted communities C_0, C_1, C_2 of size $n_c = 10$ each. In an uncalibrated DC-SBM with edge probabilities $P(A_{ij} = 1) = \min(1, p_{g_i g_j}\theta_i\theta_j)$ where $\frac{1}{n_c}\sum_{i\in C_r}\theta_i = 1$, the sum of propensity products across distinct within-community pairs in community C_r is:

$$\begin{aligned}\sum_{\substack{i<j \\ i,j\in C_r}} \theta_i\theta_j &= \frac{1}{2}\left[\left(\sum_{i\in C_r}\theta_i\right)^2 - \sum_{i\in C_r}\theta_i^2\right] \\ &= \frac{n_c(n_c-1)}{2} - \frac{n_c}{2}\mathrm{Var}_{C_r}(\theta),\end{aligned} \tag{7}$$

whereas for distinct communities $r \neq s$, $\sum_{i\in C_r, j\in C_s}\theta_i\theta_j = n_c^2$ is strictly invariant to $\mathrm{Var}_{C_r}(\theta)$. Equation (7) proves that increasing within-community degree variance $\mathrm{Var}_{C_r}(\theta)$ while holding p_{in} and p_{out} fixed systematically reduces expected within-community edges $\mathbb{E}[m_{\text{in}}]$ (further exacerbated by $\min(1,\cdot)$ clipping on hub pairs) while leaving $\mathbb{E}[m_{\text{out}}]$ unchanged, thereby inflating the empirical mixing parameter $\mu = m_{\text{out}}/(m_{\text{in}} + m_{\text{out}})$.

To eliminate both community-volume skew and mixing drift, we construct a **Separation-Calibrated DC-SBM**:

1. Within each community C_r, vertex rank $u \in \{1,\ldots,n_c\}$ receives propensity $\tilde{\theta}_u = u^{-\gamma}$ normalized so $\theta_u = n_c\tilde{\theta}_u / \sum_{v=1}^{n_c}\tilde{\theta}_v$. Every community therefore has identical mean propensity 1.0 and identical total propensity $\Theta_r = 10$.
2. Given target baseline probabilities $(p_{\text{in}}, p_{\text{out}})$, we solve via 80-step bisection for scaling parameters $(\alpha_{\text{in}}, \alpha_{\text{out}})$ such that the exact clipped edge expectations match the homogeneous SBM:

$$\mathbb{E}[m_{\text{in}}] = \sum_{r=0}^{K-1}\sum_{i<j\in C_r}\min(1, \alpha_{\text{in}}\theta_i\theta_j) = K\binom{n_c}{2}p_{\text{in}}, \tag{8}$$

$$\mathbb{E}[m_{\text{out}}] = \sum_{r<s}\sum_{i\in C_r, j\in C_s}\min(1, \alpha_{\text{out}}\theta_i\theta_j) = \binom{K}{2}n_c^2 p_{\text{out}}. \tag{9}$$

7 Experimental Design

Our experimental investigation is organized in three stages:

- **Foundational Verification Suite (Days 1–6):** Validates quadratic form identities, bridge-strength scaling (0 to 4 bridges between K_4 cliques), two-way L vs. L_{sym} partitioning on a 10-vertex asymmetric graph, three-community recovery on a 9-vertex weighted-bridge graph, and homogeneous SBM robustness across five $(p_{\text{in}}, p_{\text{out}})$ pairs.
- **Day 7 Controlled Degree-Heterogeneity Benchmark:** Evaluates $N = 30, K = 3, n_c = 10$ graphs at baseline $(p_{\text{in}}, p_{\text{out}}) = (0.65, 0.08)$ ($\mathbb{E}[m_{\text{in}}] = 87.75$, $\mathbb{E}[m_{\text{out}}] = 24.00$) across three heterogeneity regimes—**Balanced** ($\gamma = 0.00$, $\mathrm{CV}_\theta = 0.0000$), **Moderately Imbalanced** ($\gamma = 0.45$, $\mathrm{CV}_\theta = 0.3572$), and **Highly Imbalanced** ($\gamma = 0.85$, $\mathrm{CV}_\theta = 0.7427$)—over 30 matched random seeds (`1000..1029`) per condition (90 graphs total).
- **Day 8 Regime Interaction & Localization Study:** Evaluates a 5×3 factorial grid spanning five inter-community edge probabilities $p_{\text{out}} \in \{0.02, 0.06, 0.10, 0.15, 0.22\}$ at $p_{\text{in}} = 0.65$ and $\gamma \in \{0.00, 0.45, 0.85\}$ over 30 trials per cell (450 graphs total, seeds `2000..2029`), plus a 90-graph sparse vulnerability study comparing standard L_{sym} against degree-regularized L_τ ($\tau = 0.25\bar{d}$) as p_{in} drops from 0.65 to 0.32.

8 Algorithms and Implementation Details

All matrices and solvers are implemented in `src/spectral_core.py` using `numpy.linalg.eigh` (which returns eigenvalues in ascending order and orthonormal eigenvectors for real symmetric matrices). For each graph instance G and target cluster count $k = 3$, we evaluate four matched clustering pipelines sharing the exact same deterministic k-means procedure (`SpectralGraph._kmeans` with `seed=42` and `max_iter=100`):

1. **Ordinary Spectral Clustering (L):** Extracts $U_L = [u_1, u_2, u_3] \in \mathbb{R}^{n\times 3}$ from $L = D - A$ and clusters the rows of U_L.
2. **Normalized Spectral Clustering (L_{sym}, NJW):** Extracts $U_{\text{sym}} = [v_1, v_2, v_3] \in \mathbb{R}^{n\times 3}$ from $L_{\text{sym}} = D^{-1/2}LD^{-1/2}$, normalizes every non-zero row ($\|U_{\text{sym},i,:}\|_2 > 10^{-12}$) to unit Euclidean norm $Y_{i,:} = U_{\text{sym},i,:}/\|U_{\text{sym},i,:}\|_2$, and clusters the rows of Y.
3. **Ablation Control 1 (`Ord+RowNorm`):** Applies NJW spherical row normalization to the ordinary Laplacian eigenvectors U_L prior to k-means.
4. **Ablation Control 2 (`Norm-NoRowNorm`):** Clusters the unnormalized eigenvectors U_{sym} of L_{sym} directly without row normalization.
5. **Regularized Normalized Clustering (L_τ):** Replaces D with $D_\tau = D + \tau I$ ($\tau = 0.25\bar{d}$) in $L_\tau = I - D_\tau^{-1/2}AD_\tau^{-1/2}$ before NJW row normalization and k-means [5].

9 Evaluation Metrics

Partition accuracy against planted labels $g \in \{0,1,2\}^n$ is measured via the Hubert–Arabie Adjusted Rand Index (ARI $\in [-1, 1]$, where 1.0 denotes perfect recovery and 0.0 denotes chance agreement). For each trial t, we compute the paired difference:

$$\Delta\mathrm{ARI}_t = \mathrm{ARI}_{\text{norm},t} - \mathrm{ARI}_{\text{ord},t}. \tag{10}$$

To quantify eigenvector localization on nontrivial eigenvectors $j \in \{2,\ldots,k\}$ (indices `1..k-1`), we compute the mean Inverse

Participation Ratio (IPR):

$$\mathrm{IPR}(U) = \frac{1}{k-1}\sum_{j=2}^{k}\sum_{i=1}^{n} U_{ij}^4 \in \left[\frac{1}{n}, 1\right], \tag{11}$$

which equals $1/n = 0.0333$ for a perfectly delocalized vector and approaches 1.0 when an eigenvector concentrates on a single vertex. We also record the squared embedding norm $\mathrm{MinDeg}\, E(L) = \|U_{L,i_{\min},:k}\|_2^2$ on the minimum-degree vertex $i_{\min} = \arg\min_i d_i$ and the spectral threshold ratio $\lambda_3(L)/\max(d_{\min}, 1)$.

10 Results

10.1 Foundational Verification Results (Days 1–6)

Prior to Day 7, four baseline experiments established core spectral properties:

- **Bridge-Strength Scaling (`bridge_strength.py`):** Across $0,1,2,3,4$ unit bridges linking two K_4 cliques, $\lambda_2(L)$ increases strictly monotonically $(0.0000, 0.3542, 0.7639, 1.2679, 2.0000)$ alongside best Ncut $(0.0000, 0.1538, 0.2857, 0.4000, 0.5000)$, recovering the exact planted $[1..4] \mid [5..8]$ split in every case.
- **Two-Way Laplacian Comparison (`compare_laplacians.py`):** On the 10-vertex test graph with degrees $[3,2,4,5,6,4,6,4,6,6]$, zero-thresholding the ordinary Fiedler vector yields an imbalanced 7-vs-3 split with Ncut $= 0.9650$, whereas the normalized Fiedler vector yields a balanced 5-vs-5 split with Ncut $= 0.7077$.
- **Three-Community Recovery (`multi_community.py`):** On 9 vertices (three K_3 cliques joined by 0.2-weight bridges), L_{sym} yields 1 connected component, bottom eigenvalues $[-0.000, 0.029, 0.089]$, and exact cluster sizes $[3,3,3]$.
- **Homogeneous SBM Robustness (`clustering_robustness.py`):** Over 10 seeds on $N = 30, K = 3$ graphs, mean normalized ARI transitions from 1.0000 ± 0.0000 at $(p_{\mathrm{in}}, p_{\mathrm{out}}) = (0.80, 0.02)$ and 0.9706 ± 0.0621 at $(0.60, 0.10)$ to 0.1878 ± 0.2296 at $(0.40, 0.20)$, 0.0235 at $(0.30, 0.25)$, and 0.0059 at $(0.25, 0.25)$ (random baseline -0.0145).

10.2 Day 7 Controlled Degree-Heterogeneity Results

Table 1 and Figure 1 summarize the Day 7 experiment. First, propensity calibration holds realized mean degree ($\bar{d} \in [7.39, 7.41]$), within-community edges ($\bar{m}_{\mathrm{in}} \in [87.13, 87.53]$), between-community edges ($\bar{m}_{\mathrm{out}} \in [23.47, 24.03]$), and realized mixing ($\bar{\mu}_{\mathrm{emp}} \in [0.2112, 0.2164]$) invariant across all three regimes while cleanly varying realized degree dispersion from $\mathrm{CV}_d = 0.244$ (mean $d_{\min}/d_{\max} = 3.8/11.1$) to $\mathrm{CV}_d = 0.423$ ($2.5/14.9$). All 90 trials produce single-component graphs with zero isolated vertices.

Second, consistent with Hypothesis H1, ordinary spectral clustering accuracy degrades from $\mathrm{ARI}_{\mathrm{ord}} = 0.9680 \pm 0.0925$ under Balanced degrees to 0.9170 ± 0.1768 under Moderate imbalance and 0.9073 ± 0.1812 under High imbalance, whereas normalized spectral clustering achieves 0.9900 ± 0.0401, 0.9864 ± 0.0352, and 1.0000 ± 0.0000, respectively.

Third, Table 2 demonstrates that this degradation accompanies a 44.6% compression in the ordinary three-community eigengap $\lambda_4(L) - \lambda_3(L)$ (dropping from 1.3985 ± 0.6377 to 0.7753 ± 0.4152), while the normalized eigengap $\lambda_4(L_{\mathrm{sym}}) - \lambda_3(L_{\mathrm{sym}})$ remains stable ($0.3343 \to 0.3581$). Crucially, the row-normalization ablation shows that removing row normalization from L_{sym} (`Norm-NoRowNorm`) produces identical mean ARI $(0.9900, 0.9864, 1.0000)$ to full NJW clustering, whereas adding row normalization to L (`Ord+RowNorm`) only partially mitigates the loss $(0.9832, 0.9447, 0.9698)$. Thus, degree normalization of the Laplacian operator itself ($D^{-1/2}LD^{-1/2}$) is the primary driver of robustness.

10.3 Day 8 Two-Way Regime Interaction & Localization Results

Stratifying the 90 Day 7 trials into terciles of realized mixing $\mu_{\mathrm{emp}} = m_{\mathrm{out}}/(m_{\mathrm{in}} + m_{\mathrm{out}})$ revealed that under Low Mixing ($\mu_{\mathrm{emp}} \le Q_{33}$, $\bar{\mu} \approx 0.18$), ΔARI under High degree imbalance was only $+0.0113$, whereas under High Mixing ($\mu_{\mathrm{emp}} > Q_{67}$, $\bar{\mu} \approx 0.25$), ΔARI jumped to $+0.1873$. Motivated by this observation, the Day 8 factorial experiment systematically varied $p_{\mathrm{out}} \in \{0.02, 0.06, 0.10, 0.15, 0.22\}$ across $\gamma \in \{0.00, 0.45, 0.85\}$ (450 graph instances).

Table 3 and Figure 2 confirm Hypothesis H2:

- **High-Separation Regime ($p_{\mathrm{out}} = 0.02, \bar{\mu} \approx 0.06$):** Both L and L_{sym} achieve ARI $= 1.0000$ across all 30 trials even under High degree heterogeneity ($\Delta\mathrm{ARI} = 0.0000$). Here, $\lambda_3(L)/d_{\min} \le 0.2523 \ll 1$, so the three community eigenvectors of L lie far below the minimum-degree threshold.
- **Moderate-to-High Mixing Regimes ($p_{\mathrm{out}} \in [0.10, 0.22]$):** As inter-community bridges increase, $\lambda_3(L)$ rises toward $d_{\min}$. At $p_{\mathrm{out}} = 0.15$ ($\bar{\mu} \approx 0.33$), ΔARI more than doubles across degree regimes from $+0.1280$ (Balanced) to $+0.2158$ (Moderate) and $+0.2702$ (High). At $p_{\mathrm{out}} = 0.22$ ($\bar{\mu} \approx 0.42$), $\mathrm{ARI}_{\mathrm{ord}}$ collapses from 0.5672 (Balanced) to 0.3335 (High), while $\mathrm{ARI}_{\mathrm{norm}}$ remains at 0.8844, producing $\Delta\mathrm{ARI} = +0.5509$ (95% CI $[+0.462, +0.640]$, winning $30/30$ trials).

11 Statistical Analysis

Because trial-level $\Delta\mathrm{ARI}_t$ values are bounded on $[-1, 2]$ and exhibit positive skewness when $\mathrm{ARI}_{\mathrm{norm},t}$ saturates at 1.0000 (resulting in many exact ties $\Delta\mathrm{ARI}_t = 0$ alongside large positive spikes when L suffers eigenvector localization), relying solely on a parametric paired Student's t-test could be sensitive to departures from normality. We therefore report both two-sided paired t-tests with 95% confidence intervals (df $= 29$) and non-parametric two-sided Wilcoxon signed-rank tests alongside exact win/tie/loss counts ($W/T/L$).

In Day 7 (Table 1), both tests reach identical conclusions: under Balanced degrees ($\gamma = 0.00$), $\Delta\mathrm{ARI} = +0.0220$ is not statistically distinguishable from zero ($t = 1.4012$, $p = 0.1718$;

Table 1: Day 7 Controlled Benchmark ($N = 30, K = 3, n_c = 10, p_{\text{in}} = 0.65, p_{\text{out}} = 0.08$, 30 trials per regime). Expected edge counts are calibrated to $\mathbb{E}[m_{\text{in}}] = 87.75$ and $\mathbb{E}[m_{\text{out}}] = 24.00$ across all conditions. All 90 graphs are connected ($c = 1, n_{\text{iso}} = 0$).

Regime (γ)	CV_θ	Realized $\bar{d} \pm s_d$	Realized CV_d	$\bar{m}_{\text{in}}/\bar{m}_{\text{out}}$	Mix $\bar{\mu}_{\text{emp}}$	Ord ARI (Mean ± SD)	Norm ARI (Mean ± SD)	Paired ΔARI	95% CI for ΔARI	Paired p (t / Wilcox)
Balanced (0.00)	0.0000	7.41 ± 1.79	0.244 ± 0.05	87.13/24.03	0.2164	0.9680 ± 0.0925	0.9900 ± 0.0401	$+0.0220 \pm 0.0860$	$[-0.0101, +0.0541]$	0.1718 / 0.2500
Moderate (0.45)	0.3572	7.40 ± 2.50	0.340 ± 0.05	87.53/23.53	0.2113	0.9170 ± 0.1768	0.9864 ± 0.0352	$+0.0694 \pm 0.1673$	$[+0.0069, +0.1318]$	0.0307 / 0.0313
High (0.85)	0.7427	7.39 ± 3.11	0.423 ± 0.04	87.40/23.47	0.2112	0.9073 ± 0.1812	1.0000 ± 0.0000	$+0.0927 \pm 0.1812$	$[+0.0251, +0.1604]$	0.0089 / 0.0156

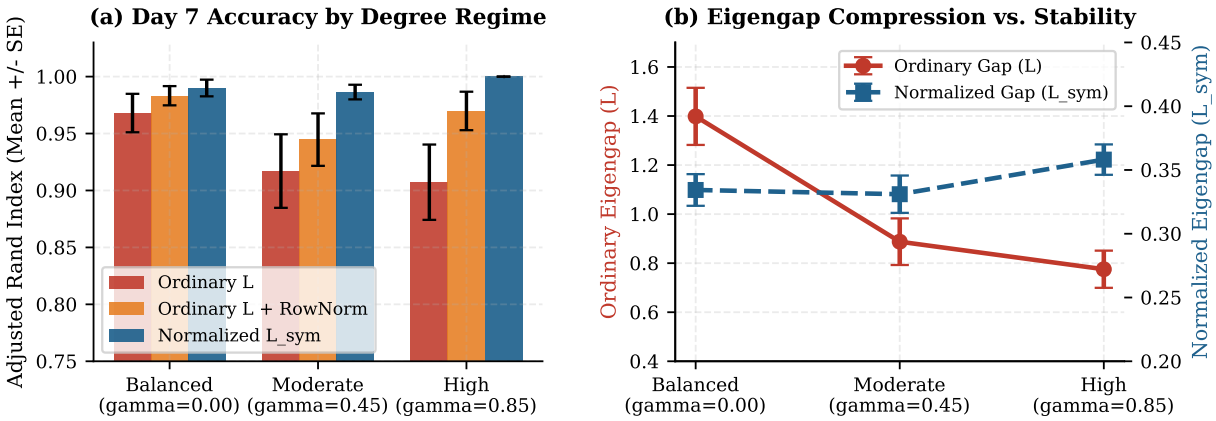


Figure 1: Day 7 experimental results ($N = 30, K = 3, p_{\text{in}} = 0.65, p_{\text{out}} = 0.08$, 30 trials per condition). (a) Mean Adjusted Rand Index (± standard error) for Ordinary L, Row-Normalized Ordinary L, and Symmetric Normalized L_{sym}. (b) Compression of the ordinary eigengap $\lambda_4(L) - \lambda_3(L)$ under increasing degree heterogeneity contrasted with the invariance of $\lambda_4(L_{\text{sym}}) - \lambda_3(L_{\text{sym}})$.

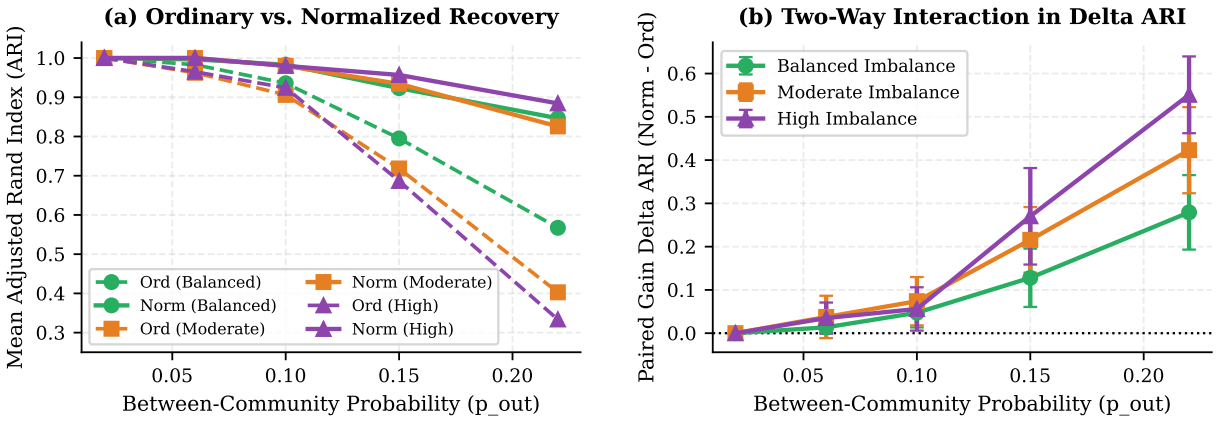


Figure 2: Two-way interaction between degree heterogeneity (γ) and inter-community edge probability (p_{out}) across 450 graphs. (a) Mean ARI trajectories for Ordinary L (dashed) and Normalized L_{sym} (solid). (b) Paired accuracy gain ΔARI with 95% confidence intervals, showing zero difference at $p_{\text{out}} = 0.02$ and super-additive divergence as $p_{\text{out}} \to 0.22$.

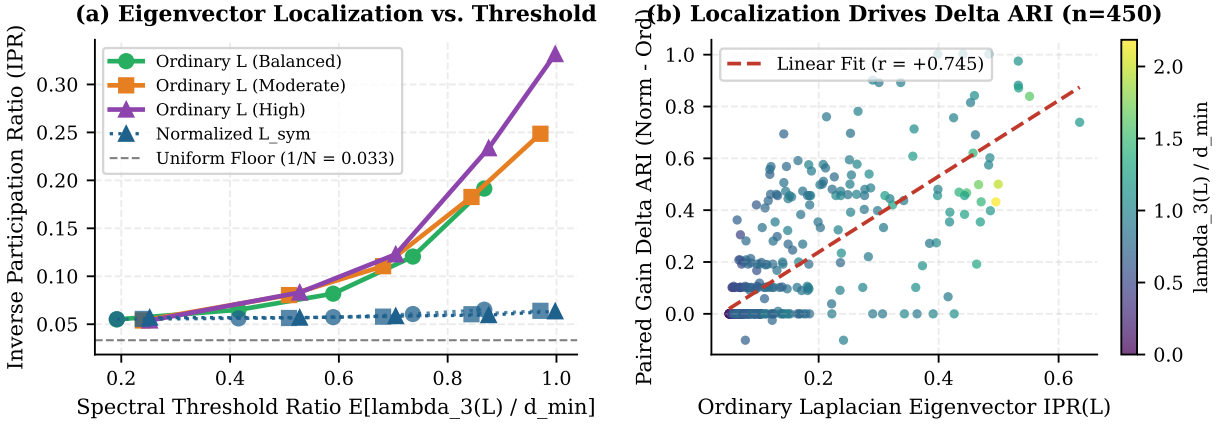


Figure 3: Mechanistic origin of ordinary spectral clustering failure. (a) Mean Inverse Participation Ratio (IPR) plotted against the spectral threshold ratio $\mathbb{E}[\lambda_3(L)/d_{\min}]$, demonstrating sharp localization of L eigenvectors as $\lambda_3(L) \to d_{\min}$ while L_{sym} remains delocalized (IPR ≈ 0.06). (b) Instance-level correlation ($r = +0.7453, n = 450$) between IPR(L) and ΔARI.

Table 2: Day 7 Eigengap ($\lambda_4 - \lambda_3$) and Row-Normalization Ablation across 30 trials per regime.

Regime	Gap L	Gap L_{sym}	Ord+RowNorm	Norm-NoRN
Balanced (0.00)	1.3985 ± 0.6377	0.3343 ± 0.0680	0.9832	0.9900
Moderate (0.45)	0.8876 ± 0.5205	0.3310 ± 0.0802	0.9447	0.9864
High (0.85)	0.7753 ± 0.4152	0.3581 ± 0.0653	0.9698	1.0000

Wilcoxon $W = 0.0, p = 0.2500$; $3/27/0$). Under Moderate imbalance ($\gamma = 0.45$), $\Delta\text{ARI} = +0.0694$ is statistically significant ($t = 2.2719, p = 0.0307$; Wilcoxon $p = 0.0313$; $6/24/0$), and under High imbalance ($\gamma = 0.85$), $\Delta\text{ARI} = +0.0927$ is highly significant ($t = 2.8040, p = 0.0089$; Wilcoxon $p = 0.0156$; $7/23/0$). Across all 90 Day 7 trials, normalized spectral clustering never underperforms ordinary spectral clustering on a single graph instance (0 losses).

Across all 450 instances in the Day 8 factorial grid, Pearson correlations confirm the theoretical perturbation links: $\text{Gap}(L)$ correlates with ARI_{ord} at $r = +0.5398$, $\text{Gap}(L_{\text{sym}})$ correlates with ARI_{norm} at $r = +0.5087$, the threshold ratio $\lambda_3(L)/d_{\min}$ correlates with ARI_{ord} at $r = -0.6557$, and eigenvector localization $\text{IPR}(L)$ correlates with ARI_{ord} at $r = -0.7494$ and with ΔARI at $r = +0.7453$ (Figure 3b).

12 Discussion

Our empirical findings reconcile two seemingly conflicting intuitions in spectral clustering. On one hand, degree heterogeneity alone does not automatically break the ordinary Laplacian L: when communities are weakly coupled ($p_{\text{out}} = 0.02$), the community cut eigenvalues $\lambda_2(L), \lambda_3(L)$ remain near zero ($\lambda_3(L)/d_{\min} \approx 0.25$), well separated from the minimum vertex degree $d_{\min}$. In this regime, the bottom k eigenvectors of L remain approximately piecewise-constant community indicators ($\text{IPR}(L) = 0.0539$), and ordinary spectral clustering achieves 100% recovery.

On the other hand, as inter-community edge density rises ($p_{\text{out}} \geq 0.08$), $\lambda_2(L)$ and $\lambda_3(L)$ scale upward with the cut size while high degree heterogeneity pushes $d_{\min}$ downward. Once $\lambda_3(L)/d_{\min}$ approaches unity (reaching 0.9977 at $p_{\text{out}} = 0.22, \gamma = 0.85$), the k-th eigenvalue enters the continuous degree spectrum $[d_{\min}, d_{\max}]$ described by von Luxburg et al. [9]. By the Courant–Fischer theorem, a unit vector concentrated on the minimum-degree vertex $i_{\min}$ achieves Rayleigh quotient $e_{i_{\min}}^T L e_{i_{\min}} = d_{\min}$, making it energetically cheaper for L to place a localized spike on $i_{\min}$ inside the bottom-k eigenspace than to represent the third community cut. Indeed, Table 3 shows that at ($p_{\text{out}} = 0.22, \gamma = 0.85$), the single lowest-degree vertex absorbs 71.04% of an entire eigenvector's squared norm (MinDeg $E(L) = 0.7104$).

13 Failure-Case Analysis

Inspection of individual failure instances reveals two distinct structural failure modes:

1. **Peripheral Singleton Splitting in Ordinary Clustering (L):** In Day 7 under Moderate imbalance ($\gamma = 0.45$), Ordinary clustering fails ($\text{ARI}_{\text{ord}} < 0.5$) on 2 of 30 trials while Normalized clustering fails on 0 (0 joint failures). In these failed trials, $u_3(L)$ localizes onto a degree-2 peripheral vertex, causing k-means to allocate an entire cluster to 1–2 peripheral nodes and merge two planted 10-node communities into a 20-node cluster (ARI ≈ 0.44).
2. **Disconnected and Pendant-Vertex Instability in Standard Normalized Clustering (L_{sym}):** To determine when L_{sym} loses its advantage, Table 4 and Figure 4 evaluate sparse high-heterogeneity regimes. When density drops to ($p_{\text{in}} = 0.32, p_{\text{out}} = 0.03, \gamma = 1.00$), the mean minimum degree falls to $\bar{d}_{\min} = 0.17$ and the mean number of connected components rises to 3.07. Because every isolated vertex or disconnected pair creates an exact zero eigenvalue in L_{sym}, the bottom $k = 3$ eigenvectors are consumed by trivial component indicators, causing standard L_{sym} accuracy to collapse to ARI $= 0.2134$ (nearly matching Ordinary L at 0.1632). Adding degree regularization $D_\tau = D + \tau I$ ($\tau = 0.25\bar{d}$) suppresses small-degree variance and restores recovery to ARI $= 0.7357$.

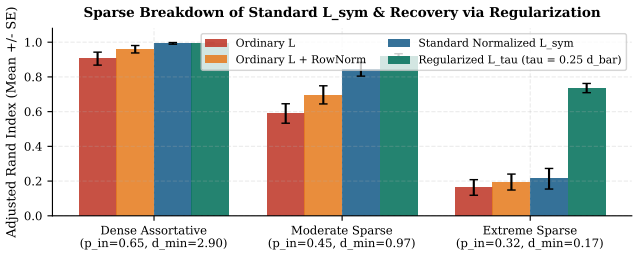


Figure 4: Breakdown of standard L_{sym} in the sparse hub-periphery regime as $d_{\min} \to 0$ and recovery via degree-regularized L_τ ($\tau = 0.25\bar{d}$).

14 Limitations

Several scope boundaries qualify our conclusions:

1. **Finite Network Scale ($N = 30$):** Our experiments focus on $N = 30, K = 3$ graphs to maintain continuity with the project's baseline benchmarks and permit exact dense eigendecompositions across 630 total graph instances. At larger scales ($N \geq 10^3$), power-law degree tails can span several orders of magnitude, sharpening the localization transition.
2. **Equal Planted Community Sizes ($n_c = 10$):** We deliberately held community sizes and expected volumes equal across all three communities to isolate vertex-level degree dispersion from community-volume imbalance.
3. **Known Cluster Count ($k = 3$):** All pipelines were supplied the true number of communities $k = 3$ rather than estimating k automatically from eigengap heuristics.

15 Conclusion

By decoupling within-community degree heterogeneity from inter-community mixing through a Separation-Calibrated DC-SBM, this study demonstrates that the advantage of symmetric normalized spectral clustering (L_{sym}) over ordinary spectral clustering (L) is governed by a fundamental interaction between degree dispersion and community separation. When $\lambda_k(L) \ll d_{\min}$, both operators recover planted communities

Table 3: Day 8 Two-Way Regime Interaction Study ($N = 30, K = 3, p_{\text{in}} = 0.65$, 30 trials per cell, 450 graphs total). Columns report empirical mixing $\bar{\mu}_{\text{emp}}$, mean ARI, paired ΔARI with 95% confidence intervals, two-sided Wilcoxon p-values, win/tie/loss counts ($W/T/L$), eigengaps, threshold ratio $\lambda_3(L)/d_{\min}$, Inverse Participation Ratio (IPR), and squared norm on the minimum-degree vertex MinDeg $E(L)$.

p_{out}	Regime	$\bar{\mu}_{\text{emp}}$	Ord ARI	Norm ARI	ΔARI	95% CI	Wilcox p	$W/T/L$	Gap(L)	Gap(L_{sym})	$\lambda_3/d_{\min}$	IPR(L)	IPR(L_{sym})	MinDeg $E(L)$
0.02	Balanced	0.0643	1.0000	1.0000	+0.0000	[+0.000, +0.000]	1.00×10^{0}	0/30/0	2.1786	0.5305	0.1918	0.0550	0.0553	0.1174
0.02	Moderate	0.0635	1.0000	1.0000	+0.0000	[+0.000, +0.000]	1.00×10^{0}	0/30/0	1.7970	0.5320	0.2396	0.0539	0.0555	0.1352
0.02	High	0.0604	1.0000	1.0000	+0.0000	[+0.000, +0.000]	1.00×10^{0}	0/30/0	1.7502	0.5672	0.2523	0.0539	0.0569	0.1255
0.06	Balanced	0.1616	0.9830	0.9966	+0.0136	[+0.000, +0.027]	1.25×10^{-1}	4/26/0	1.7103	0.4059	0.4155	0.0650	0.0561	0.1675
0.06	Moderate	0.1614	0.9625	1.0000	+0.0375	[−0.011, +0.087]	2.50×10^{-1}	3/27/0	1.3003	0.3939	0.5084	0.0804	0.0565	0.1989
0.06	High	0.1662	0.9650	1.0000	+0.0350	[−0.001, +0.071]	6.25×10^{-2}	5/25/0	1.2149	0.4263	0.5280	0.0831	0.0569	0.2549
0.10	Balanced	0.2427	0.9357	0.9830	+0.0473	[+0.013, +0.081]	7.81×10^{-3}	8/22/0	1.3318	0.2913	0.5895	0.0819	0.0574	0.2619
0.10	Moderate	0.2392	0.9061	0.9800	+0.0740	[+0.018, +0.130]	3.91×10^{-3}	9/21/0	1.0569	0.3083	0.6813	0.1105	0.0579	0.3209
0.10	High	0.2485	0.9240	0.9800	+0.0560	[+0.006, +0.106]	3.13×10^{-2}	6/24/0	0.8004	0.3228	0.7045	0.1229	0.0585	0.3273
0.15	Balanced	0.3289	0.7951	0.9232	+0.1280	[+0.061, +0.195]	7.62×10^{-4}	15/13/2	1.0105	0.1999	0.7361	0.1206	0.0607	0.3630
0.15	Moderate	0.3278	0.7185	0.9342	+0.2158	[+0.140, +0.291]	8.78×10^{-5}	20/10/0	0.7516	0.2050	0.8438	0.1827	0.0604	0.4908
0.15	High	0.3245	0.6867	0.9568	+0.2702	[+0.159, +0.382]	5.20×10^{-5}	21/8/1	0.6080	0.2342	0.8754	0.2335	0.0596	0.6048
0.22	Balanced	0.4169	0.5672	0.8465	+0.2793	[+0.193, +0.365]	1.23×10^{-5}	25/5/0	0.8025	0.1202	0.8673	0.1914	0.0652	0.4701
0.22	Moderate	0.4148	0.4028	0.8257	+0.4229	[+0.323, +0.522]	3.78×10^{-6}	28/2/0	0.5473	0.1189	0.9705	0.2486	0.0640	0.5847
0.22	High	0.4166	0.3335	0.8844	+0.5509	[+0.462, +0.640]	1.73×10^{-6}	30/0/0	0.4597	0.1375	0.9977	0.3316	0.0635	0.7104

Table 4: Sparse Peripheral Vulnerability Study (30 trials per regime). Standard L_{sym} breaks down when $d_{\min} \to 0$ and components > 1, whereas regularized L_τ ($\tau = 0.25\bar{d}$) maintains recovery.

Sparse Regime	$\bar{d}_{\min}$	Comp	Ord	Ord+RN	Norm	Reg L_τ
Dense (0.65, 0.08, $\gamma = 0.85$)	2.90	1.00	0.9053	0.9594	0.9932	0.9932
Mod. Sparse (0.45, 0.05, 0.85)	0.97	1.20	0.5894	0.6966	0.8448	0.9200
Ext. Sparse (0.32, 0.03, 1.00)	0.17	3.07	0.1632	0.1944	0.2134	0.7357

accurately. When increasing degree heterogeneity or inter-community mixing drives $\lambda_k(L)/d_{\min} \to 1$, the ordinary eigengap $\lambda_{k+1}(L) - \lambda_k(L)$ collapses and the nontrivial eigenvectors of L localize sharply onto minimum-degree peripheral vertices ($r = -0.7494$ with ARI_{ord}), causing severe clustering failures that spherical row normalization alone cannot fully repair. Conversely, in ultra-sparse hub-periphery networks where $d_{\min} \to 0$, standard L_{sym} itself degrades due to disconnected and pendant vertices, requiring degree regularization (L_τ) for reliable community detection.

16 Future Work

Promising extensions of this work include: (1) evaluating the random-walk Laplacian $L_{\text{rw}} = D^{-1}L$ and entrywise eigenvector-ratio embeddings (SCORE) under joint degree and community-size heterogeneity; (2) deriving data-driven selection rules for the regularization parameter τ in L_τ based on empirical degree quantiles; and (3) scaling the benchmark to sparse Lanczos solvers (`scipy.sparse.linalg.eigsh`) on real-world citation and social networks.

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